ECE 329 **Homework 7** **Due: Thursday, October 15, 2026, 11:59:00PM**

- Homeworks are due Thursday at 11:59:00 p.m. Late homework will not be accepted.
- Homeworks are to be turned in to Gradescope.
- Any deviation from the following steps will result in a 5 point penalty:
  - Write your name, netID, and section ONLY on the first page of your submission.
  - Start each new problem on a new page.
  - Scan the homework as a PDF (rather than taking pictures). Use a free scanning app if you do not have access to a photocopier.
  - Upload the PDF to Gradescope and tag each problem's location in the PDF.
- Each student must submit individual solutions for each homework. You may discuss homework problems with other students registered in the course, but you may not copy their solutions. If you use any source outside of class materials that we've provided, you must cite every source that you used.
- Use of homework solutions from past semesters is not allowed and is considered cheating. Copying homework solutions from another student or generative AI is considered cheating.
- Penalties for cheating on homework: a zero for the assignment on the first offense, and an F in the course on the second offense.

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Reading Assignment: Kudeki: Lectures 17-20

1.

a) I have a slab of some insulator with an unknown permittivity ϵ. To determine ϵ experimentally I go to the lab and insert the slab in between the plates of a capacitor whose plate spacing exactly matches the width of the slab. I observe that the time constant of exponential decay of the capacitor voltage in an RC circuit that I construct increases by 50% when the slab is inserted to replace the air spacing. Determine ϵ in terms of ϵ_o. Explain your reasoning carefully.

b) Repeat (a) if the slab width is only one half of the plate separation so that when the slab is inserted between the plates we still have 50% air filling.

c) I have a rod of some solid with an unknown permeability μ. To determine μ experimentally I go to the lab and insert the rod within a solenoid of many turns having a diameter exactly matching the diameter of the rod. I observe that the time constant of exponential decay of the solenoid current in an RL circuit that I construct decreases by 0.2% when the rod is inserted to replace the air core of the solenoid. Determine μ in terms of μ_o. Is the rod diamagnetic or paramagnetic? Explain.

2.

a) For current density $\mathbf{J} = (5z^2\,\hat{x} + 4x^3y\,\hat{y} + 3z(y-y_o)^2\,\hat{z})$ A/m^2, which is time independent, find the charge density $\rho(0,t)$ at the origin (0,0,0) as a function of time t, if $\rho = 0$ at that location and time $t = 0$, $y_o = 2$ m, and coordinates x, y, and z are specified in meter units. **Hint:** use the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

b) In part (a), deduce the physical units of the coefficients 5, 4, and 3 used in J_x, J_y, and J_z specifications, respectively, by applying dimensional analysis.

3.

a) Show that in a homogeneous conductor where $\mathbf{J} = \sigma\mathbf{E}$, Gauss's law $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_o}$ and the continuity equation $\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$ can be used together to derive a differential equation

$$\frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon_o}\rho = 0$$

for the charge density ρ.

b) Find the solution of the differential equation above for $t > 0$ if at $t = 0$ the charge density is $\rho(x,y,z,0) = \sin(100x)$ C/m^3 over all space.

c) According to the solution found in part (b), how long would it take for ρ to reduce to $0.01\sin(100x)$ C/m^3? Assume that $\sigma = 5.0 \times 10^7$ S/m.

d) Discuss the energetics of the process examined above: Specifically, state whether the energy per unit volume is zero or non zero at $t = 0$ and as $t \to \infty$ and state what happens to any energy stored in the quasistatic field at $t = 0$.

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4.

a) If

$$\mathbf{E} = \cos(\omega t - \beta x)\,\hat{y}\,\frac{\mathrm{V}}{\mathrm{m}},$$

$\frac{\omega}{\beta} = c$, and $\mu = \mu_o$, find the corresponding $\mathbf{H}$ by using Faraday's law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

Hint: find a suitable time varying anti-derivative for $\nabla \times \mathbf{E}$.

b) If

$$\mathbf{H} = \cos(\omega t + \beta y)\,\hat{x}\,\frac{\mathrm{A}}{\mathrm{m}},$$

$\sigma = 0$, $\frac{\omega}{\beta} = \frac{2}{3}c$ and $\epsilon = 2.25\epsilon_o$, find the corresponding $\mathbf{E}$ by using Ampere's law

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

in which $\mathbf{J} = \sigma\mathbf{E}$ and $\mathbf{D} = \epsilon\mathbf{E}$.

5. a) Show that the waveform

$$y(x,t) = f(\omega t - \beta x) + g(\omega t + \beta x)$$

satisfies the scalar wave equation

$$\frac{\partial^2 y}{\partial t^2} = v_p^2 \frac{\partial^2 y}{\partial x^2}$$

where the constant $v_p = \omega/\beta$ represents the speed of propagation of the travelling wave.

b) Is

$$\mathbf{E} = \cos^2(\omega t - \beta x)\,\hat{y}\,\frac{\mathrm{V}}{\mathrm{m}},$$

a valid solution to

$$\frac{\partial^2 \mathbf{E}}{\partial x^2} + \frac{\partial^2 \mathbf{E}}{\partial y^2} + \frac{\partial^2 \mathbf{E}}{\partial z^2} = \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

the vector wave equation? Explain mathematically and using a physical (word) description. What relationship between β, ω, μ and ϵ must hold for this solution to be valid?

c) Is

$$\mathbf{E} = \cos(\omega t)\cos(\beta x)\,\hat{y}\,\frac{\mathrm{V}}{\mathrm{m}},$$

a valid solution to the wave equation? Explain mathematically and using a physical (word) description.

d) Is

$$\mathbf{E} = \cos(\omega t - \beta_x x - \beta_y y)\,\hat{z}\,\frac{\mathrm{V}}{\mathrm{m}},$$

a valid solution to the wave equation? Explain mathematically and using a physical (word) description. What relationship between β, ω, μ and ϵ must hold for this solution to be valid?

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6. A z-polarized plane TEM wave is propagating in vacuum (i.e., $v = c \approx 3 \times 10^8$ m/s and $\eta = \eta_o \approx 120\pi\,\Omega$) in $-x$ direction. If in the wave field varies at $x = 0$ with time according to $E_z(0,t) = 2\triangle(\frac{t}{\tau})$ V/m, where $\tau = 3\,\mu$s and $\triangle(\frac{t}{\tau})$ is the unit triangle function with width τ,

   a) Determine the vector wavefield $\mathbf{E}(x,t)$ written explicitly in terms of all space and time variables x and t,

   b) Determine the accompanying wavefield $\mathbf{H}(x,t)$ in A/m units,

   c) Determine the maximum value of *Poynting vector* $\mathbf{E} \times \mathbf{H}$.

   d) What trajectory function $x = x(t)$ describes instantaneous locations of the peak of $\mathbf{E} \times \mathbf{H}$.

   e) Plot $E_z(x,t)$ vs t at $x = -1000$ m.

   f) Plot $H_y(x,t)$ vs x at $t = 20\,\mu$s.

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